



TN-IMPACT

Tamil Nadu Innovation Modernization through
Problem Solving and Advanced Creative Tech



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IDENTIFYING INDUSTRIAL CHALLENGES

PROBLEM STATEMENT NUMBER - TNI26001

PROBLEM STATEMENT TITLE -
LOW-COST, WEARABLE MEDICAL DEVICE FOR CONTINUOUS
MONITORING OF CARDIOVASCULAR PARAMETERS TO DETECT
HYPERTENSION AND STROKE RISK.

Submitted by

TEAM MEMBER 1: SRIKUMARAN S (Leader) - 240801331

TEAM MEMBER 2: VIHITH D - 240801379

TEAM MEMBER 2: SHAILESH KUMAR V - 241801256

TEAM MENTOR: Dr J SARANYA

Certificate

This report is submitted by **The NeuroNuts** consisting of students, **Srikumaran S and Vihith D**, pursuing B.E in Electronics and Communication Engineering, and **Shailesh Kumar V**, pursuing B.Tech in Artificial Intelligence and Data Science, 2nd year, IV Semester, from Rajalakshmi Engineering College under the guidance of **Dr J Saranya**, [Associate Professor] , as part of the TN-IMPACT 2026 Hackathon.

SIGNATURE

(Dr. J Saranya)

SIGNATURE

(Srikumaran S)

SIGNATURE

(Vihith D)

SIGNATURE

(Shailesh Kumar V)

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Finally, we are grateful for the learning opportunity this project provided, allowing us to explore interdisciplinary concepts spanning embedded systems, biomedical sensing, and preventive healthcare. The experience has greatly enhanced our understanding of real-world engineering challenges and reinforced the importance of designing accessible and impactful technological solutions.

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1. ABSTRACT

Cardiovascular diseases account for approximately **32%** of global deaths annually, making them the leading cause of mortality worldwide. Among these, over **1.2 billion** adults suffer from **hypertension**, while **stroke** alone results in nearly **6.5 million** deaths each year. Alarmingly, about 46% of individuals with hypertension remain undiagnosed, often recognizing their condition only after severe complications arise. This challenge is further worsened by the limited accessibility and affordability of continuous monitoring solutions, particularly in low- and middle-income communities.

Guided by the principle that *prevention is better than cure*, this project proposes an affordable wearable cardiovascular monitoring system designed to promote early awareness and proactive health management. The device is built on an **ESP32**-based platform integrating a **PPG sensor** and **motion unit** to continuously measure heart rate, heart rate variability, oxygen saturation, and trend-based blood pressure indicators.

The system further incorporates a lightweight **TinyML**-based personalization layer that **adapts baseline physiological parameters** over time using both population-level training data and individual user history, enabling more accurate deviation detection and early risk assessment. A multi-stage signal processing pipeline combined with TinyML-assisted risk classification is employed to distinguish normal physiological variations from potential hypertension or stroke risks while minimizing false alerts. Motion-aware signal handling, personalized baseline calibration, and multi-parameter risk analysis enable reliable monitoring under varying user conditions.

The wearable provides **real-time on-device feedback**, manual user modes, and an **emergency alert feature** capable of notifying a designated caregiver during critical events. Collected physiological data can also be visualized through a dedicated web dashboard, allowing users to track long-term trends and assess improvements resulting from lifestyle changes.

By combining low-cost hardware with intelligent signal processing and embedded TinyML, the proposed system supports adaptive personalized monitoring while remaining suitable for low-power wearable deployment. This approach aims to encourage early risk recognition, reduce preventable complications, and support preventive healthcare for underserved populations.

2. INTRODUCTION

Cardiovascular diseases (CVDs) remain the leading cause of mortality worldwide, accounting for approximately 32% of all global deaths annually. Among these, hypertension and stroke are the most widespread and impactful conditions. Hypertension affects more than 1.2 billion adults globally, while stroke is responsible for nearly 6.5 million deaths each year. Alarming, nearly 46% of individuals with hypertension are unaware of their condition, and many receive a diagnosis only after severe complications such as stroke, heart failure, or organ damage have already occurred.

A major challenge in managing hypertension and stroke is their often-silent progression. Hypertension can remain asymptomatic for years, allowing cardiovascular risk to accumulate unnoticed, while stroke may occur suddenly with minimal warning. Consequently, many individuals seek medical attention only after acute events, when treatment becomes more complex, costly, and less effective. Delayed diagnosis contributes not only to increased mortality but also to long-term disability, psychological distress, and reduced quality of life.

The consequences of late detection extend beyond medical impact:

- Physical: prolonged recovery, long-term medication dependence, permanent impairments
- Psychological: anxiety, depression, fear of recurrence
- Financial: emergency hospitalization costs and treatment burden, especially severe for low- and middle-income households

Cardiovascular risk is not limited to individuals with pre-existing conditions. Modern lifestyle and environmental factors contribute significantly across all age groups:

- Physically demanding occupations → chronic exhaustion and dehydration
- Sedentary office work → inactivity, stress, irregular sleep
- Students and young adults → academic pressure, poor diet, excessive screen exposure
- Elderly populations → age-related physiological decline and limited medical follow-up

These factors highlight that cardiovascular risk is widespread and influenced by both physical and psychological stressors, making early awareness essential.

Inequality in access to preventive healthcare further aggravates the problem. While urban populations with financial resources benefit from regular screening and private healthcare access, large segments of the population—particularly in countries such as India—depend primarily on episodic clinical visits or emergency care. Preventive monitoring is often neglected due to cost, infrastructure limitations, or lack of awareness, resulting in late-stage intervention and increased healthcare expenditure.

- Existing monitoring solutions also present limitations:
- Hospital-based diagnostics are periodic, not continuous
- Advanced consumer wearables (e.g., ECG-enabled devices) are often expensive
- Many systems provide raw measurements without contextual interpretation
- Cuffs, electrodes, and complex setups reduce daily usability

These challenges indicate the need for affordable, accessible, and continuous preventive monitoring systems that move beyond periodic clinical measurements. Early awareness enables individuals to adopt healthier lifestyles, manage stress, seek timely medical consultation, and reduce severe outcomes through proactive intervention.

In this context, this work proposes a low-cost wearable cardiovascular monitoring system designed to promote early risk awareness and preventive healthcare. The system integrates multi-parameter physiological sensing with motion-aware signal processing and a lightweight TinyML-based personalization layer. Traditional signal processing techniques are used to extract heart rate, heart rate variability, oxygen saturation, and trend-based blood pressure indicators, while embedded TinyML models adapt individual baseline profiles over time using both population-level training data and user-specific historical readings.

By combining personalized baseline learning with multi-parameter deviation analysis, the system enables early identification of abnormal cardiovascular trends while minimizing false alerts caused by motion or transient physiological changes. The proposed approach emphasizes affordability, usability, and on-device intelligence, bridging the gap between clinical diagnostics and continuous everyday health monitoring, particularly for underserved populations.

3. LITERATURE REVIEW

I) Wearable Technology for Cardiovascular Monitoring:

Nazir et al., in “*Wearable Technology and Its Potential Role in Cardiovascular Health Monitoring and Disease Management*”, discusses the increasing adoption of wearable devices for continuous cardiovascular assessment. Their review highlights how photoplethysmography (PPG), electrocardiography (ECG), and biosensors enable monitoring of heart rate, arrhythmias, and early cardiovascular abnormalities outside clinical environments. The authors emphasize that integrating wearable data with artificial intelligence (AI) and IoMT platforms can support predictive healthcare and personalized disease management.

Similarly, Olawade et al., in “*Integrating AI Wearables into Stroke Risk Assessment*”, describe how continuous physiological monitoring using wearables can assist in identifying early stroke risk markers such as hypertension, abnormal heart rhythms, and oxygen desaturation. These studies establish the importance of real-time monitoring for preventive cardiovascular care.

However, both works acknowledge that current commercial wearables are often expensive and inaccessible to underserved populations, limiting large-scale adoption.

II) PPG-Based Cuffless Blood Pressure Monitoring:

Chen et al., in “*A Review: Blood Pressure Monitoring Based on PPG and Circadian Rhythm*”, provides a comprehensive overview of cuffless BP estimation techniques using PPG waveform characteristics. They explain how pulse amplitude, rise time, and waveform morphology correlate with arterial pressure and how circadian rhythm influences BP variation.

Qin et al., in “*Advances in Cuffless Continuous Blood Pressure Monitoring*”, further summarize PPG-based approaches including feature extraction, pulse transit time (PTT), and regression models. Their review highlights the advantages of PPG for wearable applications due to its noninvasive nature and compact implementation. However, both papers emphasize major limitations such as motion artifacts, signal instability, skin tone variability, and the need for calibration, which significantly affect real-world accuracy.

III) Machine Learning Approaches for Blood Pressure Estimation:

Recent studies increasingly employ machine learning to improve BP prediction accuracy. Babu et al., in “*AI-Powered Framework for Non-Invasive Continuous Blood Pressure Tracking Using Smartwatch Sensors*”, proposed a CNN–LSTM architecture trained on PPG and motion data, achieving mean absolute errors of approximately 3–5 mmHg under controlled datasets.

Ara, in “*Machine Learning Models for Predicting Blood Pressure Using Wearable Sensors*”, demonstrated regression-based models using features such as pulse arrival time and HRV metrics to estimate BP. Additionally, the study “*Quality-Aware PPG-Based Blood Pressure Classification for Energy-Efficient Trustworthy BP Monitoring Devices with Reduced False Alarms*”

introduced signal quality assessment techniques to improve reliability by filtering poor-quality PPG segments prior to ML inference.

While these approaches show promising accuracy, they depend heavily on large, labeled datasets, subject-specific calibration, and computationally intensive processing. While these approaches demonstrate promising accuracy, they typically require large labeled datasets and computationally intensive architectures. However, recent advancements in TinyML and embedded optimization techniques have made it feasible to deploy lightweight machine learning models on low-power microcontrollers, enabling adaptive personalization without excessive computational overhead.

IV) Motion Artifacts and Signal Reliability:

Several studies identify motion artifacts as a primary challenge in wearable BP monitoring. The quality-aware PPG classification framework highlights how movement severely degrades signal quality and increases false alarms. To address this, researchers propose sensor fusion techniques combining PPG with accelerometer data to suppress unreliable measurements.

However, these methods introduce additional computational overhead and energy consumption, making them unsuitable for compact, battery-powered wearable systems intended for continuous use.

V) Wearables for Hypertension and Stroke Risk Assessment:

Gumprecht et al., in “Avoiding Stroke: A Continuous Monitoring Challenge”, emphasizes the importance of prolonged cardiovascular monitoring for detecting arrhythmias and stroke precursors. Olawade et al. further demonstrate that wearable-based monitoring of HR, BP trends, and oxygen saturation can provide early indicators of stroke risk.

These works suggest that trend-based physiological monitoring and early warning systems can significantly improve preventive outcomes, even without providing exact clinical BP values. VI)

Comparative Analysis of Existing Methods:

The most accurate BP estimation methods reported in literature rely on deep learning models (CNN–LSTM) or PTT-based ECG–PPG fusion. While effective in research settings, these techniques require complex hardware, extensive computation, and frequent calibration, limiting their feasibility in low-cost embedded wearables.

Commercial smartwatches similarly avoid continuous standalone BP estimation and instead provide trend-based insights or require cuff-based calibration.

In contrast, this project adopts a lightweight physiological trend-based approach. Rather than estimating absolute BP values, the system classifies BP status levels (low, normal, elevated, high) using heart rate, HRV, SpO₂, PPG trends, and motion context. This strategy prioritizes early risk awareness, low power consumption, and affordability over clinical-grade precision.

In contrast to deep CNN–LSTM architectures used in research settings, the proposed system integrates deterministic signal processing with a lightweight TinyML-based personalization layer. Rather than predicting absolute blood pressure values, the embedded model refines risk classification using extracted physiological features and adaptive baseline learning. This hybrid strategy maintains computational efficiency while enabling individualized cardiovascular risk assessment on resource-constrained wearable hardware.

VII) Research Gaps and Motivation for the Proposed System:

The reviewed literature predominantly focuses on improving BP estimation accuracy through computationally intensive ML models, with limited emphasis on affordability, embedded deployment, and accessibility. Most systems remain laboratory-based or dependent on high-end hardware.

Motivated by these gaps, this project proposes a compact, low-cost wearable integrating PPG and IMU sensing to provide continuous cardiovascular monitoring and early indicators of hypertension and stroke risk. By emphasizing frugal innovation and deployable embedded processing, the system aims to support preventive healthcare in resource-constrained environments.

The proposed approach bridges the gap between computationally intensive clinical ML models and simple threshold-based wearable systems. By leveraging TinyML for embedded personalization and combining it with motion-aware signal processing, the system aims to achieve a balance between accuracy, interpretability, affordability, and real-world deployability.

Author/Year	Method Used	Key Technique	Sensors Used	Reported Accuracy	Computational Complexity
Chen et al., 2024	PPG + Circadian Analysis	Signal morphology + circadian modeling	PPG	MAE = 6-10 mmHg	Medium
Qin et al., 2022	PPG Feature Regression	Feature extraction + ML regression	PPG	RMSE = 7-12 mmHg	High
Sivanajeyulu et al., 2023	Quality-Aware ML (CNN + LTSM)	Deep Learning + signal quality assessment	PPG	MAE = 5-7 mmHg	Very high
Ara, 2025	Wearable ML Regression	XG Boost + behavioral features	PPG + IMU	MAE = 5 mmHg	High
Multiple studies (MIMIC/UCI databases)	PTT/PAT based	Pulse transit time modeling	ECG + PPG	MAE = 6-9 mmHg	High
Commercial Smartwatches	Periodic cuff calibration	Proprietary models + calibration	PPG	Not publicly disclosed	High
Proposed System	Multiparameter risk classification	Threshold based trend analysis + personalized baseline	PPG + IMU	Categorical risk levels (No absolute BP)	Low

Table 1: Comparison of Existing Cuffless Blood Pressure Estimation Approaches

4 (A): PROBLEM STATEMENT

1. Background and Clinical Need:

Cardiovascular diseases (CVDs), particularly hypertension and stroke, remain among the leading causes of morbidity and mortality worldwide. These conditions often progress silently, with symptoms appearing only at advanced stages. Early identification of cardiovascular risk factors such as elevated heart rate, reduced heart rate variability, abnormal oxygen saturation, and rising blood pressure trends is essential for timely intervention and improved outcomes. However, traditional clinical assessments typically provide intermittent measurements, failing to capture continuous physiological changes occurring throughout daily life.

2. Limitations of Existing Monitoring Solutions:

Current health monitoring approaches present several limitations. Cuff-based blood pressure devices, while accurate, are unsuitable for continuous or ambulatory use. Commercial wearable devices offer partial cardiovascular insights but are often expensive, limiting accessibility for low- and middle-income populations. Furthermore, many advanced research systems rely on computationally intensive machine learning models and large clinical datasets, restricting their deployment on low-power wearable platforms.

In addition, practical challenges such as motion artifacts, inconsistent user compliance, and lack of personalized baseline monitoring reduce the reliability of existing solutions. As a result, early cardiovascular risks frequently go undetected, leading to delayed diagnosis and increased healthcare burden.

3. Societal and Practical Challenges:

Individuals exposed to occupational stress, sedentary lifestyles, physical labor, or age-related health decline are particularly vulnerable to cardiovascular conditions. Yet many lack access to continuous monitoring tools or regular medical checkups. This issue is especially prominent in resource-constrained settings, where preventive healthcare infrastructure is limited.

Consequently, many people become aware of their condition only after experiencing severe symptoms or complications.

4. Objective of the Proposed System:

The objective of this project is to design and develop a low-cost wearable device for continuous cardiovascular monitoring and early risk awareness. The proposed system integrates photoplethysmography (PPG) and inertial sensing to monitor heart rate, heart rate variability, blood oxygen saturation, and blood pressure status levels.

Rather than attempting clinical-grade blood pressure measurement, the device focuses on identifying abnormal physiological trends associated with hypertension and stroke risk. Motion aware signal processing, adaptive baseline calibration, and persistent deviation detection are employed to minimize false alarms and improve reliability under real-world conditions.

5. Scope and Intended Impact:

By emphasizing frugal innovation, embedded real-time processing, and user-centric design, this project aims to provide an accessible preventive healthcare tool that promotes early awareness and encourages timely medical consultation. The system is intended not as a replacement for professional diagnosis, but as a continuous monitoring aid that empowers individuals—particularly those in underserved communities—to proactively manage their cardiovascular health.

Furthermore, conventional wearable monitoring systems largely rely on fixed thresholds and generalized population averages, which often fail to account for individual physiological variability and changing user conditions. This limitation leads to inaccurate risk assessment and frequent false alerts. To address this gap, the proposed system integrates a lightweight TinyML-based personalization layer that continuously adapts individual baseline parameters using both population-trained models and historical user data. This enables more reliable deviation detection, context-aware risk classification, and improved early warning accuracy while remaining suitable for deployment on resource-constrained embedded hardware.

4 (B): DESIGN METHODOLOGY

4.1) System Overview:

The proposed wearable device is designed for continuous cardiovascular monitoring and early risk awareness of hypertension and stroke. The system integrates optical sensing, motion detection, embedded processing, and user interaction into a compact, low-cost platform.

Key Functional Flow:

- PPG and motion signals are acquired continuously
- Motion artifacts are suppressed using IMU data
- Physiological parameters (HR, HRV, SpO₂) are extracted
- Blood pressure is estimated as categorical status (low/normal/elevated/high)
- Personalized baselines are maintained
- Multi-parameter fusion determines cardiovascular risk
- Results are displayed locally and prepared for dashboard visualization

Rather than attempting clinical-grade BP estimation, the device focuses on trend-based risk detection, enabling preventive awareness while remaining computationally lightweight.

4.2) TinyML-Based Personalized Risk Analysis Module:

In addition to conventional signal processing and parameter extraction, the proposed system incorporates a lightweight TinyML inference module deployed on the ESP32 for personalized cardiovascular risk assessment. Raw sensor signals are first filtered and conditioned to remove motion artifacts and noise. Physiological parameters such as heart rate, heart rate variability, SpO₂, and trend-based blood pressure indicators are computed using deterministic algorithms. These parameters, along with contextual inputs including age, gender, temperature, and motion state, are combined into a feature vector.

The TinyML model, pre-trained offline using population-level datasets and deployed on-device for inference, analyzes deviations from dynamically learned user baselines to classify cardiovascular risk into categories such as normal, hypertension risk, and stroke risk. Baseline profiles are continuously updated using historical user data, enabling adaptive personalization over time while maintaining low computational overhead suitable for embedded hardware.

4.3) Hardware Architecture:

The hardware design consists of five primary subsystems:

- Processing Unit:

An ESP32-C3 microcontroller serves as the central controller, selected for its low power consumption, compact size, and sufficient processing capability for real-time signal analysis and decision logic. It coordinates all sensor communication, executes embedded algorithms, and manages user interaction.

- Physiological Sensing Unit:

A MAX30102 PPG sensor provides red and infrared signals for heart rate, HRV, and SpO₂ estimation. The sensor was chosen for its low cost, small footprint, and suitability for wearable applications.

- Motion Sensing Unit:

An MPU6050 IMU supplies accelerometer data used for detecting motion, classifying activity states, and suppressing unreliable PPG measurements during movement.

- User Interface:

A compact OLED display presents real-time vitals, operating modes, and alerts. A multifunction push button enables navigation, manual mode selection, and emergency activation.

- Power Management:

The system is powered by a rechargeable lithium-ion battery, managed by a TP4056 charging module and a 3.3 V LDO regulator. Firmware-level power optimization (idle modes and sensor duty cycling) extends battery life.

All subsystems communicate via I²C and GPIO, forming a modular architecture that supports future scalability.

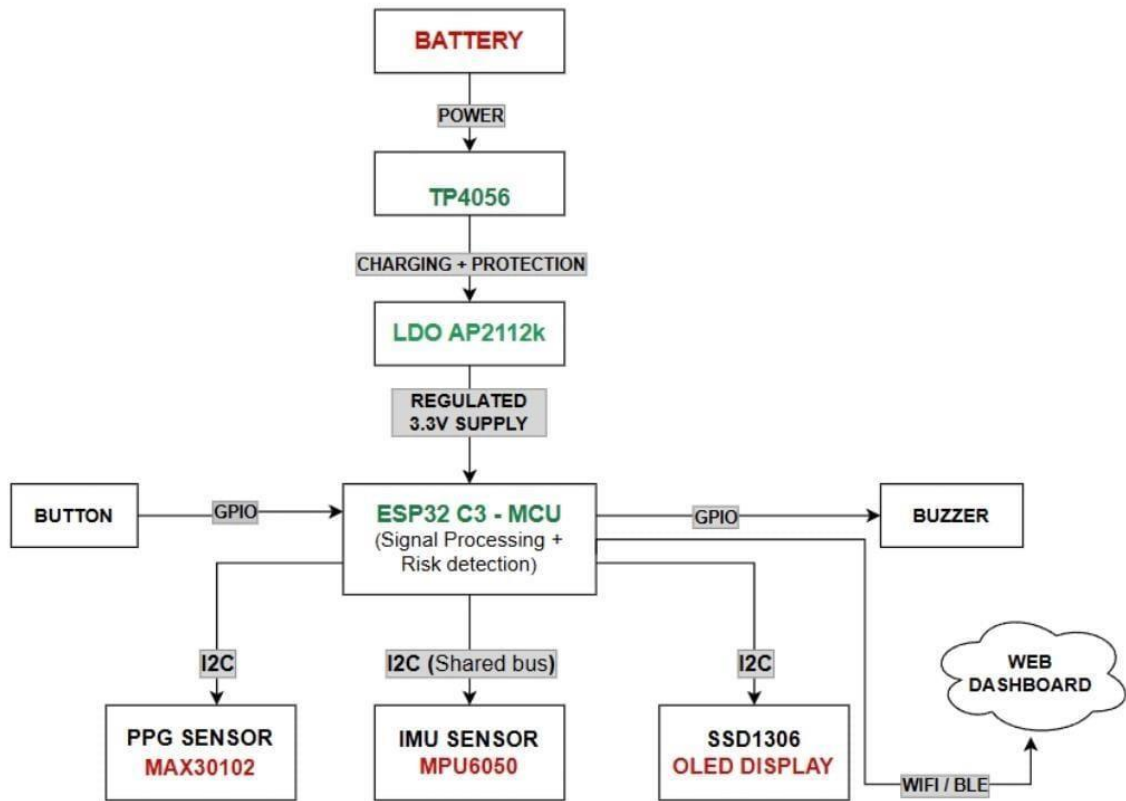


Fig 1: *Hardware Block Diagram of the Proposed Wearable System*

Component	Model	Function	Interface	Features offered
Microcontroller	ESP32-C3	Central Processing	I ² C / GPIO	Low cost, Wi-Fi capable
PPG Sensor	MAX30102	HR, HRV, SpO ₂ acquisition	I ² C	Compact, widely used
IMU	MPU6050	Motion detection	I ² C	Reliable, low-cost
Display	SSD1306 OLED	Parameter display	I ² C	Low power
Buzzer	Piezo	Alert signaling	GPIO	Simple alert system

Table 2: *Hardware Components used in the Proposed Wearable device*

4.4) Functional Features and Operating Modes:

The device supports both automatic and manual operation to adapt to real-world usage.

○ Operating Modes:

- Idle Mode: minimal motion; reduced sampling for power saving
- Sleep Mode: prolonged inactivity; lowest power operation
- Passive Movement Mode: light daily activity; motion-aware filtering enabled
- Active Movement Mode: high motion; PPG gating and recovery logic activated

Activity classification is performed using accelerometer magnitude thresholds.

○ Manual Control: A single push button allows: -

- parameter navigation (push for <1s)
- mode selection (push for 1-2s)
- emergency activation via long press (push for >3s)

This ensures accessibility even without smartphone interaction.

4.5) Signal Acquisition and Motion-Aware Processing:

PPG signals are digitally filtered to remove noise and baseline drift. IMU data is continuously analyzed to detect motion artifacts.

During excessive movement:

- physiological measurements are temporarily paused
- corrupted data is discarded

After physical activity, a recovery phase is enforced before recalculating baselines, preventing exercise-induced changes from generating false alerts.

This motion-aware strategy significantly improves reliability under everyday conditions.

4.6) Adaptive Baseline Calibration:

Initial resting measurements establish personalized baseline values for HR, HRV, SpO₂, and PPG amplitude.

These baselines are updated gradually using moving averages. Risk evaluation is delayed following exertion until parameters stabilize, ensuring alerts reflect sustained physiological changes rather than temporary fluctuations.

4.7) Cardiovascular Parameter Estimation:

- Heart Rate: derived from peak detection in filtered PPG
- HRV: computed from inter-beat intervals
- SpO₂: estimated using the red/infrared ratio-of-ratios method

Instead of numeric blood pressure, the system classifies BP into:

- ➔ Low
- ➔ Normal
- ➔ Elevated
- ➔ High

using combined trends of HR, HRV, SpO₂, and PPG amplitude. This avoids unreliable cuffless BP values while preserving clinical relevance.

Parameter classification:	LOW	NORMAL	ELEVATED	HIGH
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Eg: Classification of a parameter showing different levels as compared to the baseline.

4.8) Risk Evaluation and Alert Classification Framework:

Each parameter is categorized relative to baseline and clinical reference ranges.

Multi-Parameter Fusion Logic:

Alerts are generated only when multiple abnormalities persist simultaneously, minimizing false positives.

Progressive Alert Levels:

- Level 1 – Informational: mild deviation (visual warning only)
- Level 2 – Warning: sustained abnormality
- Level 3 – Emergency: multiple critical parameters

Hypertension risk is inferred from sustained HR elevation, reduced HRV, and elevated BP status. Stroke risk indicators include severe combined abnormalities and oxygen desaturation.

Individual parameter deviations are shown with warning symbols, while disease-level alerts require composite conditions.

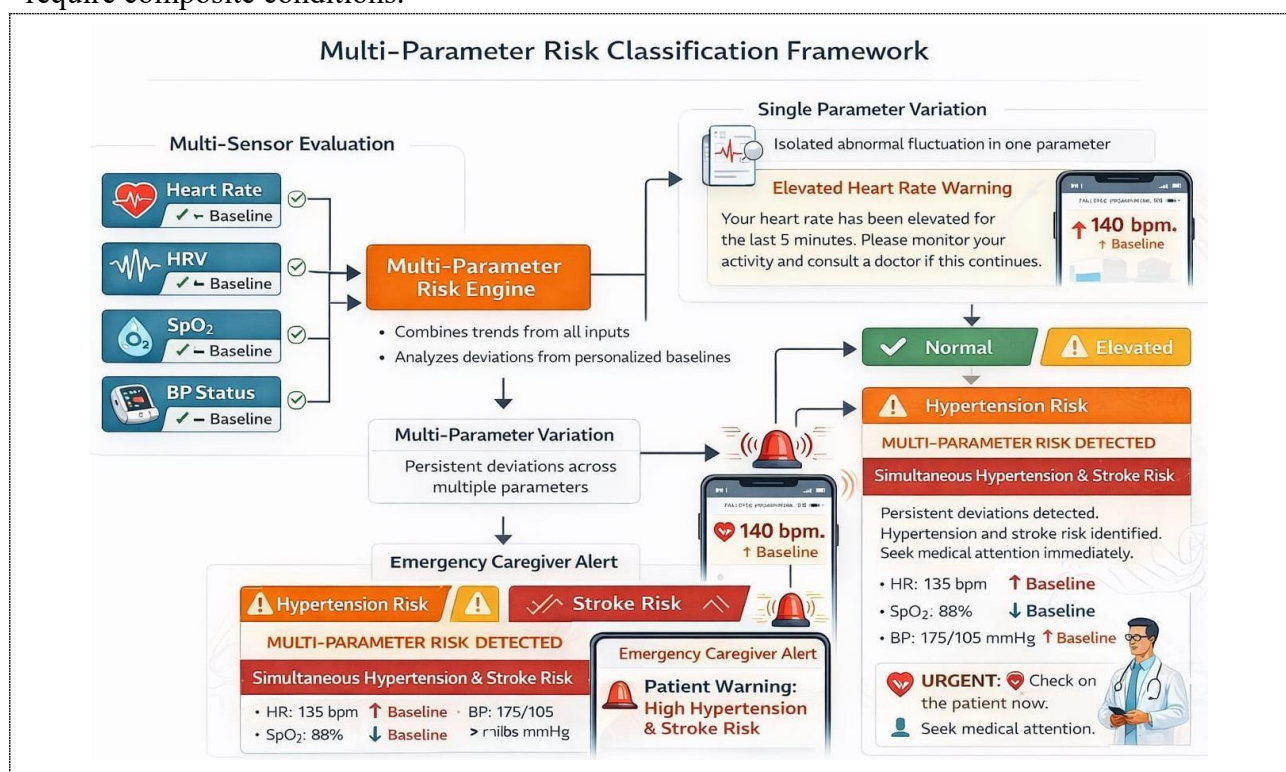


Fig 2: Multi-Parameter Risk Classification Framework showing single-parameter warnings, multi-parameter hypertension/stroke risk detection, and caregiver emergency alert triggering.

Parameter	Personalized Baseline Range	Elevated / Reduced	High-Risk Deviation	Interpretation
Heart Rate (BPM)	60 - 90	90 – 110 / <60	>110 / <50	Elevated HR may indicate stress or exertion; sustained high/low values suggest cardiovascular strain
Heart Rate Variability (ms)	40 - 80	25 - 40	<25	Reduced HRV suggests increased stress or poor autonomic regulation
SpO ₂ (%)	95 - 100	90 - 94	<90	Lower oxygen saturation may indicate respiratory or circulatory issues
Estimated BP level	Normal	Elevated	High (Hypertension Risk)	Derived from combined HR, HRV, SpO ₂ trends and PPG amplitude
Motion level	Rest / Light movement	Moderate Activity	Sudden Impact/Fall	Used to suppress unreliable readings and enable recovery phase

Table 3: (Example) Parameter Classification for an Adult Male User: (Age 20-45, Moderate activity level)

NOTE: The values presented in this table represent an example classification framework for an adult male user with moderate activity levels. In the actual system, personalized baseline calibration is performed for each individual during initial usage. Risk evaluation is conducted relative to the user's own physiological baseline rather than generalized age- or gender-based averages, ensuring improved accuracy and reduced false alerts across diverse users.

4.9) User Interaction and Emergency Handling:

The OLED displays real-time vitals, activity state, and alerts. Emergency mode can be triggered either manually or automatically upon detecting critical physiological patterns.

Emergency actions include:

- local buzzer alarm
- visual notification
- preparation for wireless caregiver alerts (future extension)

4.10) System Workflow Summary:

Overall operation follows:

Sensor acquisition → motion filtering → baseline calibration → parameter extraction → risk evaluation → user notification

This integrated workflow enables continuous cardiovascular monitoring while maintaining low power consumption, affordability, and usability.

The hybrid architecture combines conventional signal processing with embedded machine learning to improve robustness under real-world conditions. Motion-aware filtering ensures reliable physiological measurements, while the TinyML layer enables personalized deviation analysis based on both population-trained models and individual historical data. This approach reduces false alarms, enhances early risk detection, and maintains low power consumption, making the system suitable for continuous wearable deployment.

5. DEVELOPMENT OF PROTOTYPE

5.1) Prototype Overview:

A functional prototype of the proposed wearable device was developed to validate system feasibility and demonstrate real-time physiological monitoring. The prototype integrates the ESP32-C3 microcontroller, MAX30102 PPG sensor, MPU6050 IMU, OLED display, and power management circuitry on a compact prototyping board.

The primary objective of the prototype was to demonstrate acquisition of cardiovascular parameters, motion-aware operation, user interaction through display and button input, and embedded risk analysis. The prototype also serves as a proof-of-concept for future miniaturization and enclosure design.

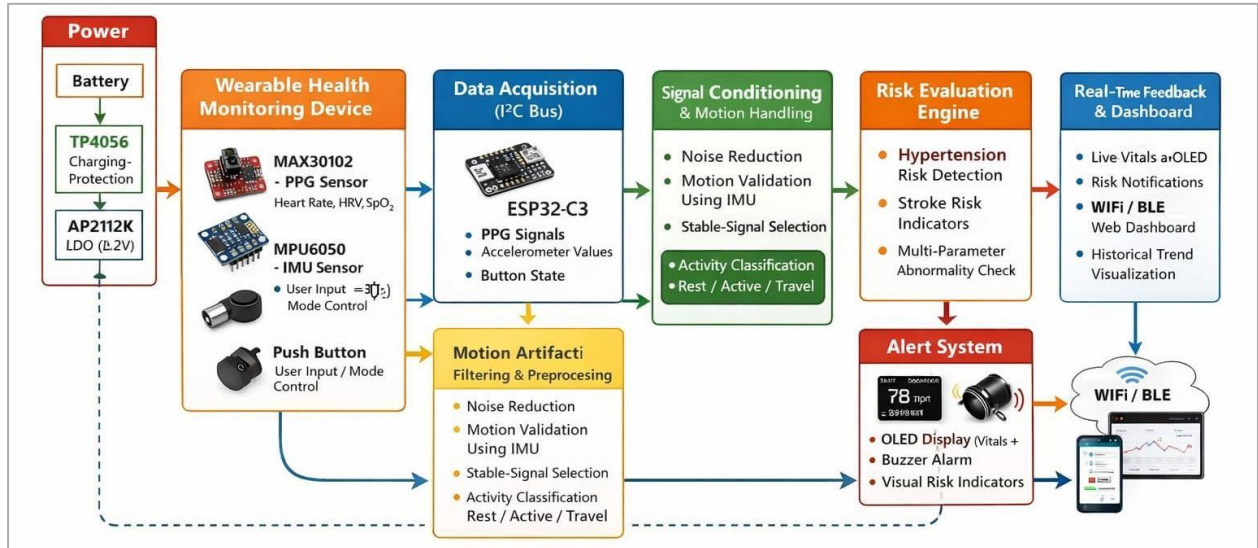


Fig 3: System Architecture of the Wearable Cardiovascular Monitoring Platform.

5.2) Hardware Assembly:

5.2.1: Component Integration

All electronic components were mounted on a perfboard and interconnected using soldered jumper wires. The ESP32-C3 was positioned centrally to minimize routing complexity. The MAX30102 PPG sensor, MPU6050 IMU, and OLED display were connected to the microcontroller through a shared I²C communication bus.

A common 3.3 V power rail was distributed to all peripherals via a low-dropout voltage regulator, while a common ground plane ensured signal stability. Care was taken to reduce wire length and avoid electromagnetic interference, particularly for the PPG sensor.

5.2.2: Power System Implementation

The prototype is powered by a rechargeable lithium-ion battery. A TP4056 charging module enables USB-based charging and battery protection, while a 3.3 V LDO regulator supplies stable voltage to the microcontroller and sensors.

The system automatically switches between external USB power during charging and battery power during standalone operation. Firmware-level power optimization, including idle states and reduced sensor sampling during inactivity, was implemented to extend operational time.

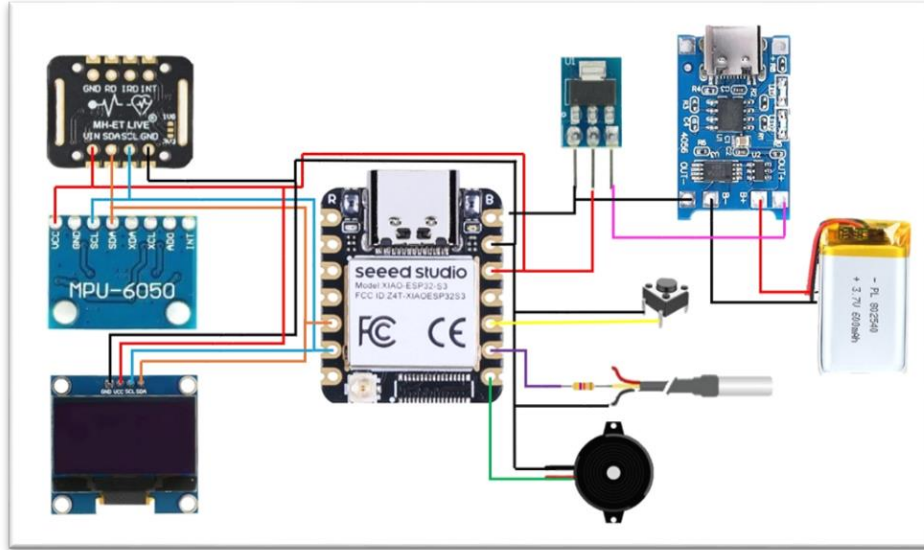


Fig 4: *Hardware Layout of the Prototype Setup*

5.3) Firmware Development:

Firmware was developed using the Arduino framework for the ESP32-C3. The software architecture follows a modular structure, separating sensor drivers, signal processing, parameter estimation, risk evaluation, and user interface control.

Key firmware components include:

- PPG acquisition and filtering routines
- IMU-based motion detection
- Heart rate, HRV, and SpO₂ computation
- Blood pressure status classification
- Adaptive baseline calibration
- OLED display handling
- Button-based navigation and control

This modular design simplifies debugging and allows future extensions, such as wireless data transmission and cloud integration.

5.4) Signal Processing and Parameter Extraction:

Raw PPG signals from the MAX30102 are digitally filtered to remove noise and baseline drift. Heart rate is calculated using peak detection, while HRV is derived from inter-beat intervals. SpO₂ is estimated using the ratio-of-ratios method based on red and infrared signals.

Instead of calculating absolute blood pressure values, the system evaluates BP status levels by combining trends in HR, HRV, SpO₂, and PPG amplitude. This approach enables meaningful cardiovascular risk assessment without relying on computationally intensive cuffless BP models.

5.5) Motion-Aware Operation:

Accelerometer data from the MPU6050 is continuously monitored to detect user movement. During periods of excessive motion, physiological measurements are temporarily suspended to prevent corrupted readings. Following physical activity, the system enters a recovery phase before recalculating baselines.

This motion-aware logic significantly reduces false alerts caused by transient physiological changes during movement.

5.6) User Interface Implementation:

The OLED display provides real-time visualization of physiological parameters, operating modes, and warning indicators. A multifunction push button allows users to navigate display screens, manually select activity modes, and activate emergency alerts through predefined press durations.

This simple interface ensures accessibility while maintaining device compactness.

5.7) Prototype Demonstration:

The prototype successfully demonstrated real-time acquisition of heart rate and activity status. During resting conditions, stable heart rate values were displayed, while motion detection triggered appropriate mode of transitions and recovery behavior.

Warning indicators were observed during sustained deviations from baseline values, validating the multi-parameter risk evaluation framework. Although full dashboard integration and emergency notifications are reserved for future development, the current prototype establishes the feasibility of embedded cardiovascular monitoring and risk awareness on a low-cost wearable platform.

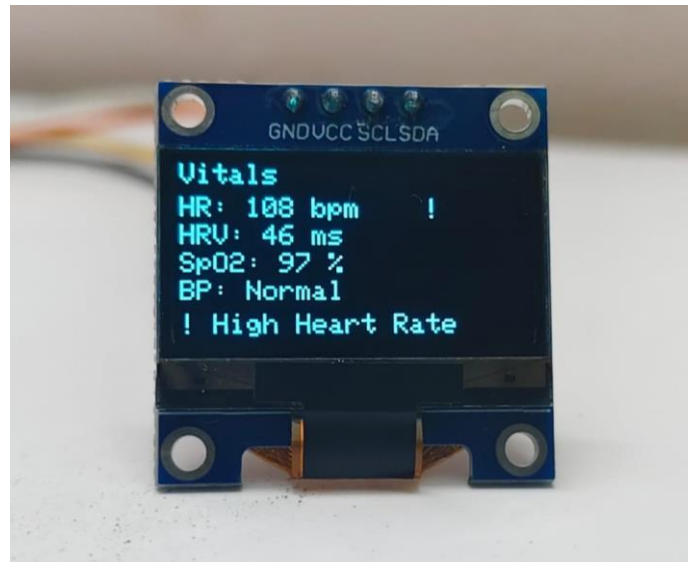


Fig 5: Sample display of Cardiovascular Parameters on the Proposed Wearable Device.

In parallel with conventional embedded firmware development, provision has been made for integrating a lightweight TinyML inference module within the ESP32 processing pipeline. While core physiological parameters are computed using deterministic signal processing techniques, the TinyML component operates at a higher level to refine personalized baseline adaptation and assist risk classification. This design ensures that machine learning remains a supportive enhancement rather than a replacement for primary signal processing, preserving system reliability and computational efficiency.

6. DESIGN VALIDATION & PROTOTYPE IMPLEMENTATION

6.1) Prototype Assembly and Hardware Integration:

The wearable prototype was assembled based on the finalized system architecture, integrating all major hardware components into a single embedded platform. The prototype includes the ESP32-C3 microcontroller, MAX30102 photoplethysmography (PPG) sensor, MPU6050 inertial measurement unit (IMU), SSD1306 OLED display, push-button input, buzzer module, and power management circuitry.

All components were mounted on a prototyping board and interconnected using jumper wiring. The PPG sensor, IMU, and OLED display communicate with the microcontroller via a shared I²C bus, enabling synchronized acquisition of physiological and motion data. A centralized 3.3 V power rail distributes regulated voltage to all modules.

Images of the assembled prototype will be included to demonstrate physical realization and component-level integration. This stage confirms that the proposed architecture is practically implementable within a compact wearable form factor.

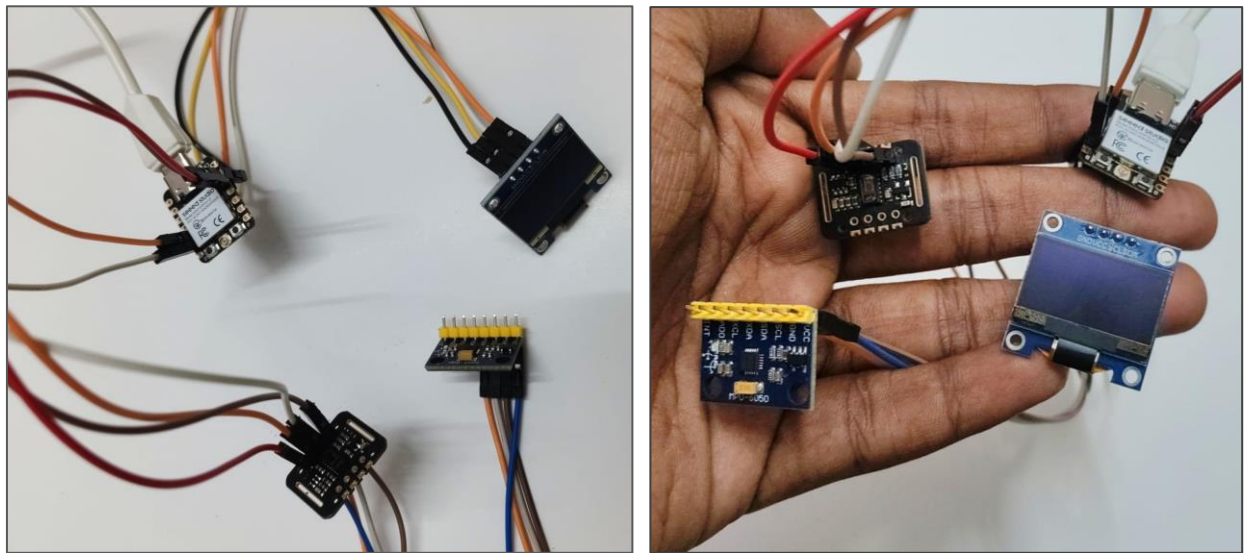


Fig 6: *Hardware assembly of the sensors and microcontroller used for the prototype.*

Figure above shows the integration of the following sensors & display with ESP32 C3 module:

- MAX30102
- MPU6500
- SSD1306 OLED Display

6.2) Embedded System Framework:

The ESP32-C3 serves as the central processing unit, coordinating sensor communication, signal preprocessing, operational mode management, risk evaluation, and user interaction. The firmware is designed using a modular architecture, separating individual subsystems such as sensor drivers, signal processing routines, risk classification logic, and display control. This structure allows independent testing of each module and supports future expansion.

The embedded software framework supports:

- Continuous acquisition of PPG and IMU data
- Motion-aware preprocessing and noise suppression
- Computation of cardiovascular parameters (HR, HRV, SpO₂)
- Classification of blood pressure status levels
- Adaptive baseline calibration
- Mode selection through both manual input and automatic motion detection
- Alert generation and user notification

Although full real-time parameter visualization is planned for upcoming development phases, the firmware structure has been established to accommodate all functional requirements.

6.3) Operational Modes and User Interaction:

The system is designed to support multiple operating modes to adapt to varying user activities:

- Sleep Mode
- Idle Mode
- Passive Movement Mode
- Active Mode
- Emergency Mode

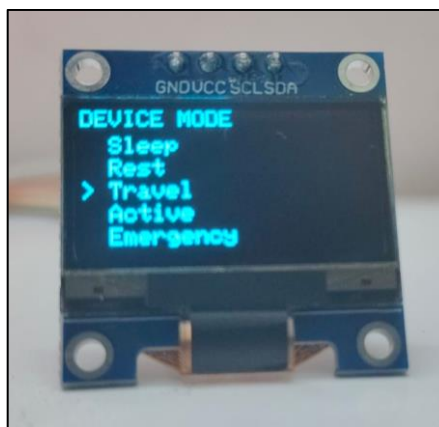


Fig 7: *Various Operating modes shown on the OLED display with automatic and manual selection options.*

Mode transitions can be initiated manually through a push-button interface or automatically using IMU-derived motion patterns. This dual-mode control ensures flexibility and accessibility for users of different age groups and technical familiarity.

The OLED display provides local visualization of physiological parameters, operating status, and alert notifications. Button-based navigation allows users to scroll through display screens, select modes, and trigger emergency actions.

6.4) Motion-Aware Signal Processing:

To ensure reliable measurements during daily activities, motion data from the MPU6050 is continuously monitored. During periods of excessive movement, physiological readings are temporarily gated to prevent motion artifacts from corrupting PPG signals. Following physical activity, a recovery phase is enforced before recalculating baseline values. This prevents transient exercise-induced changes from being interpreted as cardiovascular risk, improving system robustness under real-world conditions.

6.5) Risk Evaluation and Emergency Alert Implementation:

The system employs a multi-parameter risk evaluation framework, combining heart trends, HRV, SpO₂, motion patterns, and estimated blood pressure status.

Physiological parameters are categorized into predefined ranges such as normal, elevated, and high. Single-parameter deviations result in warning notifications, while sustained abnormalities across multiple parameters activate high-risk alerts.

An emergency alert mechanism is incorporated to support rapid response. Emergency mode can be triggered either manually by the user or automatically when critical physiological patterns are detected. The system provides local buzzer alarms and visual notifications, with planned future integration of remote alerts to a designated caregiver or family member through wireless communication.



Fig 8: Display message during Stroke risk detection.

6.6) Planned Data Visualization and Remote Monitoring:

In addition to on-device feedback, the wearable is designed to transmit collected data to a web based dashboard via Wi-Fi or Bluetooth. The dashboard enables:

- Real-time monitoring
- Historical trend visualization
- Health status tracking
- Remote caregiver access

This feature supports long-term health awareness and allows users to observe changes over time, reinforcing preventive healthcare behavior.

6.7) Future Development Phases:

While the physical prototype assembly has been completed, full functional deployment—including continuous sensor readings, dashboard connectivity, battery-powered operation, and emergency communication—will be implemented in subsequent development stages.

Planned enhancements include:

- Compact PCB-based design
- Integrated rechargeable battery system
- Power optimization strategies
- Multi-user data collection
- Algorithm refinement
- Extended validation

These future steps will transform the current proof-of-concept prototype into a fully operational wearable platform suitable for practical deployment.

7. RESULTS AND DISCUSSION

7.1) Prototype Development Status:

The project has progressed to physical prototype assembly and embedded system framework implementation. Core hardware components—including the ESP32-C3 microcontroller, MAX30102 PPG sensor, MPU6050 IMU, OLED display, push button, and alert buzzer—have been integrated according to the proposed system architecture.

The firmware framework supports modular sensing, motion detection, operational modes, and cardiovascular risk evaluation logic. Although full real-time dashboard connectivity and extended physiological visualization are planned for subsequent development phases, the current prototype demonstrates the feasibility of integrating multi-sensor cardiovascular monitoring within a compact wearable platform.

7.2) System Feasibility and Risk Framework:

The assembled prototype demonstrates that low-cost sensors combined with embedded processing can support continuous cardiovascular monitoring. The system architecture enables multi-parameter evaluation using heart rate, HRV, SpO₂, motion context, and categorical blood pressure trends.

Rather than relying on absolute cuff-based blood pressure estimation, the device classifies blood pressure into relative risk categories. Combined with motion-aware signal processing and adaptive baseline calibration, this multi-parameter framework is designed to reduce false alerts while providing meaningful early risk awareness.

A lightweight TinyML personalization layer is incorporated to refine deviation-based risk assessment using contextual and historical user data. Instead of replacing deterministic signal processing, the ML component operates as an adaptive enhancement layer, enabling individualized baseline adjustment while maintaining computational feasibility for resource-constrained embedded hardware.

Compared to conventional ML-centric approaches that require large datasets and high computational resources, the proposed hybrid architecture emphasizes lightweight processing, affordability, and deployability within wearable environments.

7.3) Preventive Healthcare Outcomes:

Beyond technical feasibility, the proposed system aims to support preventive healthcare by enabling continuous self-monitoring and early risk identification. By providing users with real-time physiological feedback and long-term trend visualization through a web dashboard, the device encourages health awareness and proactive lifestyle management.

Continuous monitoring allows individuals to recognize stress patterns, detect cardiovascular indicators early, and observe lifestyle-related changes, supporting corrective actions such as improving physical activity, managing stress, and seeking timely medical consultation. This reduces dependence on frequent clinical visits while empowering users to actively participate in their own health management.

The system also enhances preparedness for potential medical, physical, and financial challenges by issuing early warnings of cardiovascular risk, allowing users and caregivers to respond before conditions escalate into critical events.

7.4) Limitations and Future Improvements:

The current implementation is limited by incomplete sensor activation, absence of large-scale user testing, and pending battery integration. Blood pressure estimation is restricted to categorical risk levels, and clinical validation has not yet been conducted.

Future development will focus on compact PCB design, full sensor deployment, multi-user data collection, dashboard enhancements, emergency communication features, and further refinement of TinyML-assisted personalization for improved accuracy and adaptability.

7.5) Discussion:

Despite its preliminary development stage, the project demonstrates the practicality of a motion-aware, multi-parameter wearable system using frugal hardware and embedded intelligence. The results highlight the potential of accessible wearable technology to bridge gaps in preventive healthcare by enabling continuous monitoring, early risk awareness, and user-centric health insights.

With further refinement and validation, the system has the potential to evolve into a scalable platform supporting affordable cardiovascular health management, particularly for underserved communities.

8. KEY FEATURES AND INNOVATION

8.0) Novelty of the Proposed System

The proposed wearable introduces a practical, embedded approach to cardiovascular risk monitoring that prioritizes preventive awareness, affordability, and real-world reliability. The key novel aspects of the system are:

- **Cuffless Blood Pressure Risk Classification with lightweight TinyML Assistance**

Blood pressure is categorized into risk levels (low, normal, elevated, high) using combined trends of HR, HRV, SpO₂, and PPG amplitude, avoiding unreliable absolute BP estimation while leveraging lightweight TinyML models for adaptive personalization and improved risk sensitivity.

- **Motion-Aware Sensing with Recovery-Phase Logic**

IMU-assisted activity detection suppresses corrupted PPG signals during movement and enforces a recovery phase after exertion, preventing exercise-induced physiological changes from generating false alerts.

- **Adaptive Personalized Baseline Monitoring**

Physiological parameters are evaluated relative to individual resting baselines rather than generic population thresholds, reducing false positives and improving sensitivity across diverse users.

- **Multi-Parameter Disease-Level Risk Evaluation**

Single-parameter deviations generate warnings, while hypertension or stroke risk is triggered only when sustained abnormalities occur simultaneously across multiple indicators.

- **Frugal Embedded Architecture for Preventive Healthcare**

Low-cost sensors and lightweight embedded processing enable standalone operation without cloud dependence, supporting deployment in resource-constrained environments.

In addition, a lightweight TinyML layer supports personalized baseline adaptation and multi-parameter risk inference directly on-device, enabling improved individual sensitivity without dependence on cloud processing.

8.1) Multi-Parameter Cardiovascular Monitoring:

Unlike basic wearable devices that rely on a single physiological parameter, the proposed system continuously monitors multiple cardiovascular indicators, including heart rate (HR), heart rate variability (HRV), oxygen saturation (SpO₂), and blood pressure status levels. By integrating these parameters, the device provides a more comprehensive representation of cardiovascular health and enables early identification of abnormal physiological trends.

These extracted parameters also serve as input features for the embedded TinyML model to support adaptive baseline learning and personalized risk assessment.

8.2) Motion-Aware Intelligent Sensing:

A key innovation of the system is its motion-aware sensing framework. Using data from the inertial measurement unit (IMU), the device distinguishes between resting, passive movement, and active physical activity. During excessive motion, unreliable PPG measurements are suppressed, and a recovery phase is enforced following exertion. This prevents exercise-induced physiological changes from triggering false alerts and significantly improves reliability under real-world conditions.

8.3) Risk-Level Blood Pressure Classification:

Instead of estimating absolute blood pressure values—which is unreliable without cuff-based calibration—the system classifies blood pressure into categorical status levels (low, normal, elevated, high). These levels are derived from combined trends in HR, HRV, SpO₂, and PPG amplitude. This approach enables meaningful cardiovascular risk awareness while avoiding computationally intensive absolute BP estimation models, with TinyML further refining classification using personalized deviation patterns.

8.4) Adaptive Personalized Baselines:

The device incorporates adaptive baseline calibration to account for individual physiological differences. Initial resting measurements establish personalized reference values, which are gradually updated over time. A TinyML-based personalization layer assists this process by learning individual deviation trends from historical data, enabling more accurate baseline convergence across diverse users. Risk detection is performed relative to these personalized baselines, reducing false alarms, and improving sensitivity across diverse users.

8.5) Dual Mode Operation:

The system supports both automatic and manual operating modes. Automatic mode dynamically adjusts sensing behavior based on detected motion, while manual mode allows users to explicitly select activity states through a push-button interface. This dual-mode operation enhances usability and ensures accessibility for users with varying technical familiarity.

8.6) Emergency Alert Capability:

An emergency mode is implemented to support rapid response during critical events. Emergency alerts can be triggered manually by the user or automatically upon detecting severe multiparameter abnormalities. The system provides local audio-visual alerts, with planned future integration of remote caregiver notifications.

8.7) Frugal and User-Centric Design:

The wearable is designed using low-cost sensors, open-source hardware, and embedded processing to ensure affordability and scalability. A compact form factor, simple user interface, and minimal external dependencies make the device suitable for deployment in resource constrained environments, emphasizing preventive healthcare accessibility.

8.8) Integrated Dashboard for Long-Term Monitoring:

To support long-term health awareness, the system includes a web-based dashboard that allows users to visualize historical trends of physiological parameters. This feature enables users to track progress, identify recurring abnormalities, and make informed lifestyle adjustments, reinforcing the project's focus on prevention rather than cure.

Feature	Conventional Wearables	Proposed Wearable device
Primary Focus	Fitness tracking & periodic health metrics	Continuous cardiovascular risk awareness
Parameters Monitored	Typically HR & SpO ₂	HR, HRV, SpO ₂ , BP status classification, Motion context
Blood Pressure Measurement	Numeric estimation using ML with cuff calibration	Risk-level classification using multi-parameter trends
Motion Handling	Limited artifact suppression	IMU-assisted motion-aware filtering and recovery phase
Risk Evaluation	Single-parameter threshold alerts	Multi-parameter fusion-based risk engine
Personalization	Minimal or generic thresholds	Adaptive personalized baseline calibration
Emergency Response	Limited manual SOS (only in premium devices)	Manual + Automatic multiparameter emergency triggering
Dashboard Access	Proprietary ecosystem	Custom web-based dashboard for trend tracking
Hardware Cost Orientation	Premium consumer pricing	Frugal, low-cost embedded design

Target Population	Urban, higher-income users	Accessible solution for resource constrained communities
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Table 4: *Comparison between Conventional Wearables and Proposed Wearable Device.*

9. SCALABILITY AND FEASIBILITY

9.1) Technical Scalability:

The proposed system is designed with modular hardware and firmware architecture, enabling straightforward scalability. Sensor modules, power systems, and communication interfaces can be upgraded or replaced without altering the core processing framework. The ESP32-C3 microcontroller supports wireless connectivity, allowing future integration with cloud platforms, mobile applications, and caregiver notification systems. The modular firmware design also supports integration of lightweight TinyML inference models for adaptive baseline learning and on-device risk classification without significant increase in computational overhead.

From a production perspective, the use of widely available components such as the MAX30102, MPU6050, and ESP32 ensure supply chain reliability and simplifies large-scale manufacturing. Transitioning from perfboard prototyping to a custom PCB will significantly reduce size, power consumption, and assembly complexity, making the device suitable for mass deployment.

9.2) Practical Feasibility:

The system emphasizes embedded processing and lightweight TinyML-assisted algorithms, enabling adaptive personalization while avoiding dependence on computationally intensive models or continuous cloud connectivity. This enables reliable standalone operation and low power consumption, which are critical for wearable applications.

By focusing on blood pressure status classification rather than absolute BP estimation, the design avoids calibration-heavy and resource-intensive methods while still providing clinically meaningful risk awareness. Motion-aware sensing, adaptive baselines, and multi-parameter fusion further enhance robustness in everyday environments.

The simple OLED interface and single-button control make the device accessible to users with limited technical experience, including elderly individuals. The overall design prioritizes ease of use, affordability, and minimal maintenance.

9.3) Deployment Potential in Resource-Constrained Settings:

A major objective of this project is to support preventive healthcare in underserved communities. The low-cost hardware design, rechargeable battery operation, and local processing capability allow deployment in regions with limited medical infrastructure. Unlike premium wearable devices, the proposed system does not rely on proprietary ecosystems or expensive subscriptions.

Community health workers and primary care providers could utilize the device for preliminary cardiovascular screening, while individuals can use it for personal health awareness and early risk identification.

Components	Specification/Model	Approximate Cost
ESP32-C3 Dev Board	Seeed Studio	~ 500 - 700
Pulse Oximeter & Heart Rate Sensor	MX30102	~ 120 - 250
Temperature Sensor	DS18B20 / LM35	~ 80 - 150
Gyroscope Sensor	MPU6050	~ 120 - 180
OLED display	0.96" SSD1306 I2C Display	~ 100 - 200
Buzzer	Alert indication	~ 10 - 30
Push Button	User control	~ 5 - 10
Li-Po Battery	3,7V 650mAh	~ 450 - 700
TP4056 Charging Module	Battery charging + protection	~ 30 - 60
Voltage Regulator	3.3V LDO	~ 20 - 50
Passive Components	Resistors, Wire, Capacitors, Perfboard	~ 100 - 150

Table 5: *Specification of the Hardware Components used and their approximate costs.*

9.4) Business Model and Adoption Strategy:

A cost-effective production model can be achieved through bulk component sourcing and PCB based manufacturing. The device can be offered as a standalone wearable with optional digital services, such as cloud-based dashboards and caregiver alert subscriptions.

Potential adoption pathways include:

- Direct-to-consumer sales for personal health monitoring
- Partnerships with healthcare providers for community screening programs
- Integration with public health initiatives targeting hypertension and stroke prevention

By emphasizing preventive care rather than diagnostic replacement, the system aligns with emerging healthcare trends focused on early intervention and remote monitoring.

9.5) Future Expansion:

Future enhancements may include clinical validation studies, mobile application integration, and expanded TinyML-based personalization for improved baseline adaptation and multi-user risk modeling. Additional developments may focus on caregiver notification systems, enhanced

emergency response mechanisms, and large-scale deployment studies across varied activity levels and demographic groups. These extensions can be introduced incrementally without altering the core hardware platform, ensuring long-term scalability and adaptability.

10. CONCLUSION

This project presented the design and development framework of a low-cost wearable system aimed at continuous cardiovascular monitoring and early risk awareness of hypertension and stroke. By integrating photoplethysmography, inertial sensing, and embedded processing, the proposed device offers a multi-parameter approach to monitoring heart rate, heart rate variability, oxygen saturation, and blood pressure status levels under real-world conditions.

Unlike conventional wearable devices that rely heavily on computationally intensive models or proprietary platforms, this work emphasizes frugal innovation through affordable hardware components, motion-aware sensing, adaptive baseline calibration, and lightweight risk classification. A TinyML-assisted personalization layer enables baseline adaptation and deviation-based risk evaluation while maintaining low power consumption and standalone operation. The system combines physiological indicators with contextual motion data to reduce false alerts and provide meaningful early warnings rather than attempting precise clinical diagnosis.

The physical prototype assembly validates the feasibility of integrating multiple sensing modules into a compact wearable architecture. While full functional deployment—including live sensor visualization, dashboard connectivity, and emergency alerting—is planned for subsequent development phases, the current implementation establishes a strong foundation for future refinement and validation.

Beyond technical contributions, the proposed system highlights the importance of preventive healthcare by enabling continuous self-monitoring, promoting health awareness, and supporting timely medical consultation. The inclusion of a web-based dashboard further enhances long-term health tracking and empowers users to make informed lifestyle decisions.

With planned enhancements such as compact PCB design, extended user testing, algorithm optimization, and expanded TinyML personalization, the system has the potential to evolve into a scalable and accessible platform for cardiovascular risk awareness. Ultimately, this work demonstrates that thoughtfully designed low-cost wearable technology can play a meaningful role in improving preventive healthcare, particularly in resource-constrained communities.

11. REFERENCES

[1] World Health Organization (WHO), “*Hypertension – Key Facts and Global Prevalence*,” WHO Fact Sheet, 2024.

Available: <https://www.who.int/news-room/fact-sheets/detail/hypertension>

[2] A. Nazir, A. Nazir, M. S. W. Jamal, and S. U. R. Sadiq, “*Wearable Technology and Its Potential Role in Cardiovascular Health Monitoring and Disease Management*,” Health Science Reports, Wiley, 2025.

[3] K. Kario, “*Management of Hypertension in the Digital Era: Small Wearable Monitoring Devices for Remote Blood Pressure Monitoring*,” Hypertension, vol. 76, no. 3, 2020.

[4] D. B. Olawade et al., “*Integrating AI-Driven Wearable Devices and Biometric Data into Stroke Risk Assessment: A Review of Opportunities and Challenges*,” Clinical Neurology and Neurosurgery, Elsevier, 2025.

[5] J. Ara, “*Machine-Learning Models for Predicting Blood Pressure and Cardiac Function Using Wearable Sensor Data*,” International Journal of Scientific Interdisciplinary Research, vol. 6, no. 2, 2025.

[6] J. Gumprecht, M. Domek, and A. M. Hill, “*Avoiding Stroke: A Continuous Monitoring Challenge*,” Cerebrovascular Diseases, Karger, 2020.

[7] American Heart Association, “*Heart disease and Stroke Statistics — 2026 Update*,” Circulation, 2026.

Available: <https://professional.heart.org>

[8] *Advances in Cuffless Continuous Blood Pressure Monitoring Using PPG Signals, Review Article, 2021–2025 literature survey.*